One Physical World, Many Mathematical Descriptions: What Makes the Difference?

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Abstract

An equation, a coordinate grid and a measured event can all be mathematical without playing the same part in a world. What determines which relations constitute that world? This paper proposes four questions: what an element does, which changes preserve that role, what breaks if it is removed, and what observations distinguish the alternatives. Examples from quantum theory and geometry show how a change of notation can preserve the physics while a change of structure alters it. The distinction helps Matheism explain how many mathematical descriptions can refer to one physical reality.

1. The question mathematical monism must answer

A coordinate grid and the geometry it describes are both mathematical. They need not play the same role. Change the numbers assigned to events while preserving all distances and coincidences, and the physical predictions remain the same. Change the geometry itself, and clock comparisons or light propagation may change. The distinction concerns relations and representation, not a boundary between mathematical objects and nonmathematical matter.

Tegmark's mathematical universe proposal [1] and ontic structural realism [2,3] motivate this question, but do not become contradictory merely because a theory admits redundant descriptions. A proponent of mathematical realism can already distinguish isomorphic descriptions from different structures. The present criterion develops that distinction for the Matheism programme. It does not claim to refute Tegmark by observing that Cartesian and spherical coordinates describe the same space.

The question is: relative to a specified world model, which mathematical relations carry its physical content, and which choices merely express those relations? An answer must allow equivalent formulations. A symbol's disappearance from one formula does not show that its structural role disappeared from the theory.

2. Closure, consistency and specification

In formal logic, the deductive closure of a theory T contains the statements derivable from T. Deleting an axiom from a consistent classical theory cannot make the remaining axioms inconsistent: every model of the original theory still satisfies the remaining axioms. Accordingly, closure does not mean that

removing any statement creates a contradiction. Nor does it mean that every object is derivable from every other object.

This distinction matters even in an elementary example. A theory with an independent statement P admits several possibilities before P is supplied. Adding P can narrow those possibilities. Removing P restores alternatives; it does not destroy consistency. The earlier formulation of the closure criterion confused loss of determination with contradiction.

For the present physical proposal, a world model consists of an admissible structure, its relevant relations and any history or boundary data needed to specify the case under discussion. A partial model may constrain many admissible histories. A fully specified history selects one such case. Both partial and complete specifications can be consistent. Calling the former incomplete does not prove that it has no completion.

I use physical closure for an intended complete account of a world's internal relations. This is a proposed explanatory requirement, not the standard logical closure operator or a claim of mutual derivability. If an account omits an interaction that affects its predictions, it is incomplete for that purpose. If two descriptions differ only by an accepted representational equivalence, they need not describe two worlds.

For a model class M, write m ~ n when the theory identifies m and n as physically equivalent. Structural membership is assessed at the level of the relevant relations preserved under that equivalence. Merely defining a quotient M/~ does not establish which equivalences are physically justified. That remains a question for the theory and its evidence.

3. What an action principle supplies

An action principle constrains admissible histories through stationarity under specified variations. In familiar classical field theories this yields equations of motion. Boundary terms, allowed variations, constraints and boundary or initial data matter. A stationary action is not always a minimum, and an action need not select a unique history.

Consider a free particle with action S[x] = integral of m(dx/dt)^2/2 over time. Its equation is $d^2x/dt^2 = 0$. Every trajectory $x(t) = a + bt$ satisfies that equation. The action alone supplies neither a nor b. Specifying suitable endpoints selects a line in this simple example; more general boundary problems can have several solutions or none. This elementary calculation blocks the inference from an internal action to a uniquely selected universe.

A mathematical world can include both its dynamics and its boundary data without an outside agent choosing either. But that ontological claim must not be confused with a derivation of the data from the dynamics. A simplicity prior may compare complete descriptions including their boundaries; it does not remove the need to identify what is being compared.

An action is also not a prerequisite for mathematical structure or for every possible dynamics. A finite transition system has a precise update rule whether or not a useful variational formulation has been provided. Conversely, a block description can contain dynamical relations without itself changing in a further time. Neither a universal action requirement nor a ban on static descriptions follows from monism.

Noether's first theorem connects suitable continuous variational symmetries with conservation laws under its hypotheses [4]. Local gauge symmetries also involve Noether's second theorem and differential identities. These results do not make every coordinate convention a physical field or guarantee a conserved global energy in every spacetime.

4. Four questions for structural membership

The following questions are a method of analysis. They are not four independently established necessary and sufficient conditions for physical existence.

Role. What relations or predictions does the element help specify? A boundary condition or fixed parameter can matter even when it is not varied in an action.

Equivalence. Which changes preserve the physical content? Distinguish a relabeling from a change to the relations being labeled. Identify the transformation acting on the whole model.

Removal. Does removal change predictions, leave the model underspecified, or permit an equivalent reformulation? These are different outcomes. An inconsistent reformulation is a fourth possibility, not the automatic consequence of omission.

Discrimination. First ask whether the alternatives differ logically, whether they are equivalent representations, or whether they predict different observations. Logical equivalence and inconsistency are mathematical questions. Different physical predictions invite empirical discrimination; identical predictions do not. In the latter case, an ontological argument can still be assessed a priori and through explanatory economy.

The questions keep an indispensability argument attached to its actual scope. Something can be indispensable to one notation, another thing to a class of models, and something else to every admissible account of the observations. Moving between those claims requires an argument.

5. Worked cases

5.1 Complex amplitudes and real representations

Complex arithmetic has a straightforward real representation: a + ib corresponds to the pair (a,b), and multiplication by i corresponds to J(a,b) = (−b,a), with $J^2 = -I$. Replacing complex scalars by paired real components can retain

the relevant structure. Erasing the letter i therefore does not erase the role it performed. This algebraic example is already enough to distinguish representation from removal.

Renou and collaborators [5] compared complex quantum theory with a real Hilbert space analogue in a network with independently prepared sources and specified composition rules. Chen and collaborators [6] reported experimental correlations incompatible with the tested real formalism. The assumptions about composition and independence belong to the result; the experiments do not establish that every formulation using real numbers must fail.

A 2026 preprint by Hoffreumon and Woods [7] explicitly distinguishes product state independence from operational independence and claims an equivalence for finite protocols under the latter condition. That is a recent theoretical claim, not a settled verdict adopted here. It reinforces the need to state the model assumptions rather than announce that experiments have made a numeral physically real.

For the closure proposal, the useful question is which transformations, correlations and composition rules an equivalent representation must preserve. The evidence does not by itself decide between instrumentalism and mathematical monism.

5.2 Coordinates and geometry

A coordinate change replaces a chart and changes tensor components according to their transformation laws. The geometric object represented need not change. An active diffeomorphism acts on fields; physical equivalence in general relativity concerns the appropriate transformation of the whole configuration, subject to boundary conditions. A metric field is not simply unchanged under every active diffeomorphism.

Thus the object of interest is not the numerical list g_mu_nu in one chart. It is the geometric and relational content of a spacetime model, with its stated equivalences. Clock comparisons provide one way of probing that content. A coordinate grid written in a notebook is also a physical pattern, but its marks are not additional spacetime geometry merely because they label that geometry.

Using a smooth manifold in this argument specifies a mathematical model. Whether a physical continuum is fundamental or emerges from a discrete account remains a separate question.

5.3 Gauge structure and holonomy

In electromagnetism, changing A_mu to A_mu + partial_mu lambda preserves the field strength. In appropriate geometries, phase information around a loop can matter even when the local field strength vanishes along the accessible path, as in the Aharonov and Bohm analysis [8]. The example shows why discarding a representation must preserve its global content as well as its local derivatives.

For nonabelian gauge theories, the field strength transforms covariantly rather than remaining componentwise invariant. A holonomy generally transforms by conjugation; gauge invariant information can be extracted through suitable traces or conjugacy classes. These distinctions prevent the overly simple rule that potentials are unreal while every field strength or holonomy is individually invariant.

The criterion asks what survives an equivalent reformulation of the complete gauge structure. It does not replace gauge theory with a list of preferred symbols.

6. All mathematics and a particular world

Matheism's claim that reality is mathematical does not imply that every mathematical object is an additional constituent of this particular world. A map, the mapped territory and an imagined territory are different structures or structural roles. The physical inscription exists where it is written. That does not put the represented mountains there.

The further claim that all consistent structures are realized is an ontological premise of the wider programme. Structural membership within one model neither proves nor refutes that premise. It explains how the premise can coexist with a distinction between our world's relations and descriptions of other relations. The distinction is internal to mathematics rather than a return to two substances.

A story usually leaves its laws and history underspecified. Some stories also contain contradictions under a fixed interpretation. These cases must be separated. There is no theorem here that Harry Potter has no consistent completion: imposing our chemistry on a story whose stipulated laws differ already adds a premise. An inconsistent specification has no model under its fixed classical meanings; its text can nevertheless exist as a consistent inscription.

The same care applies to philosophical zombies. If consciousness is identical with specified organization, duplicating all that organization while deleting experience contradicts the identity thesis. The closure criterion does not independently establish that identity. The consciousness papers must supply their argument rather than count the disputed thesis twice.

7. Limits and objections

Is this merely structural realism? It belongs to that family. Its proposed contribution is a practical separation of membership, representation, specification and realization across the companion manuscripts. It does not claim that earlier structural realists failed to notice coordinates or gauge redundancy.

Is the criterion circular? It is conditional on a model and its equivalences. That is appropriate for analysing a theory already under discussion. It would

be circular to present the resulting classification as an independent proof that the theory is true. Empirical comparison and philosophical argument retain separate work to do.

Does incompleteness prohibit closure? Deductive closure and completeness are different. A theory can contain all its consequences while leaving some sentences undecided. Gödel's results concern appropriate consistent, effectively axiomatized theories capable of expressing enough arithmetic. They do not show that every finite physical description is inadequate for every purpose, or that a mathematical structure cannot exist. Conversely, invoking a total structure does not give us an effective procedure deciding all its truths.

Does the proposal derive physical realization? No. It distinguishes roles within an account of realization. To derive a particular universe, one would still need admissibility conditions, an account of dynamics and boundaries, and whatever measure is used to compare observer locations. A name for that combined structure cannot do the work of specifying it.

8. Conclusion

Both a world and its representations can be mathematical. Their roles differ. Relabeling coordinates can preserve every distance, while changing the geometry changes what clocks and rulers measure.

The four questions proposed here make that distinction concrete: identify a role, test what preserves it, remove it and compare the consequences. This helps specify what constitutes a particular world without assuming that every mathematical object is part of that world.

References

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